

## CSE 260 Lab Report 07

Name : Arif Bin Islam

ID : 22101623

Group : 05

Section : 22

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Group  
Members :

|                 |          |
|-----------------|----------|
| Arif Bin Islam  | 22101623 |
| Umme Hunna      | 22101565 |
| Jannatul Somiya | 22101698 |
| Mithila Farzana | 22101709 |

Name of the experiment: Design a circuit that outputs 2's complement of a 3 bit number using encoder & decoder.

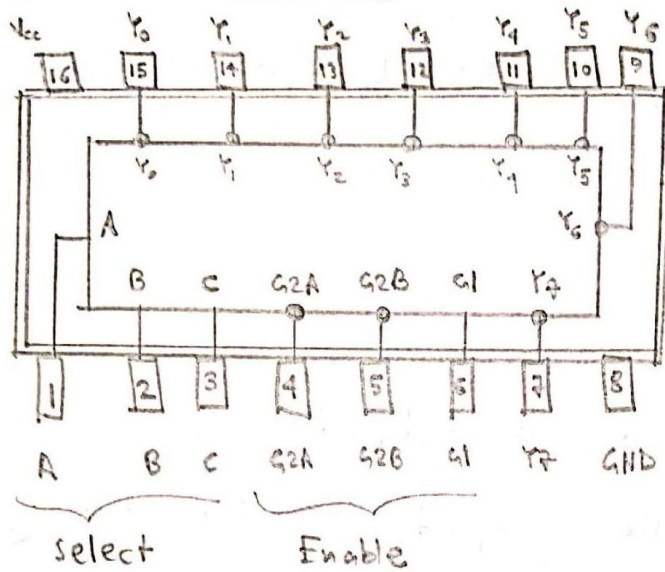
Objective: To get an output which is 2's complement of the input 3 bit binary number using one encoder and one decoder.

To gain experience and to get a better understanding of the encoder and decoder.

Required Components and Equipments:

- I. Encoder IC.
- II. Decoder IC.
- III. Wires.

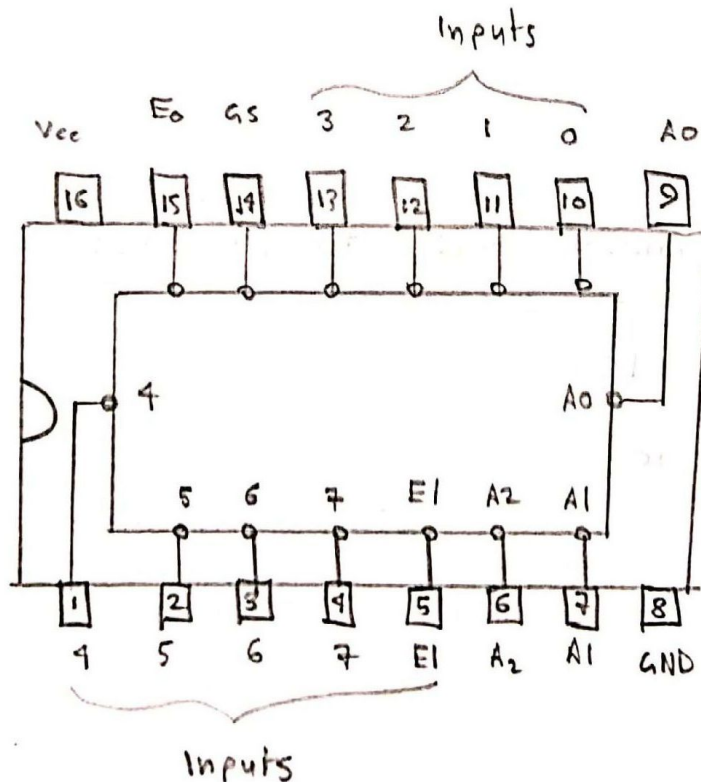
## Experimental Setup :



All y's are outputs

(8 outputs)

## 74138 - Decoder



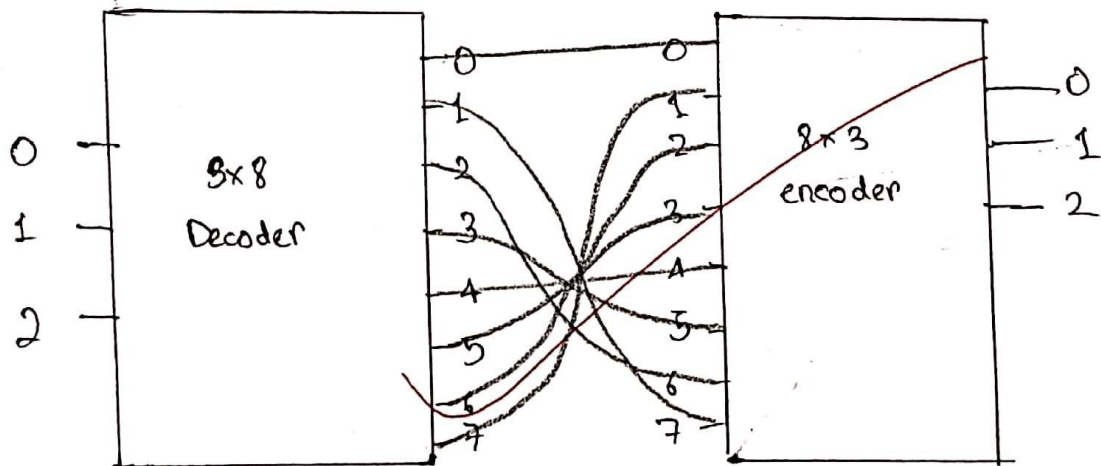
A<sub>0</sub>, A<sub>1</sub>, A<sub>2</sub>, E<sub>0</sub>, G<sub>5</sub> - Outputs

## 74148 - Encoder

## Truth table

| <u>Input</u> |   |   | <u>Active Low Output</u> |                |                |
|--------------|---|---|--------------------------|----------------|----------------|
| C            | B | A | D <sub>2</sub>           | D <sub>1</sub> | D <sub>0</sub> |
| 0            | 0 | 0 | 1                        | 1              | 1              |
| 0            | 0 | 1 | 0                        | 0              | 0              |
| 0            | 1 | 0 | 0                        | 0              | 1              |
| 0            | 1 | 1 | 0                        | 1              | 0              |
| 1            | 0 | 0 | 0                        | 1              | 1              |
| 1            | 0 | 1 | 1                        | 0              | 0              |
| 1            | 1 | 0 | 1                        | 0              | 1              |
| 1            | 1 | 1 | 1                        | 1              | 0              |

(a). Circuit diagram



Using a decoder and encoder,  
we found out the two's complement  
of a three bit binary number.

Atkins  
20.11.22

⑥. Yes, we can implement a code converter with encoder and decoder.

Decoder and Encoder circuits are used to convert one type of input ~~an~~ encoding to a different output encoding.

A  $3 \times 8$  decoder decodes a 3 bit binary number and gives an output according to the input binary number.

Also, an encoder encodes a decimal's equivalent binary number to a 3 bit binary number.

So, that's how we can implement a code converter using an encoder ~~of~~ or an decoder.

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